



Progress on Defect Predictor

By: Huy Nguyen and Edward Feng



Implementation

- Fixed calculate_recency()
- Normalized each parameter
 - MinMaxScaler()
- Fixed recency score logic
 - $1 - \text{normalized recency score}$
- Update weight distribution
- Update README.md
- Test different sample repos provided.

Revisit Parameters in the Formula

- Recency (Date committed): 25% -> 22.5%
- Frequency (How often the files are changed): 35% -> 30%
- Number of lines modified 20% -> 15%
 - Lines added + lines deleted
- How many lines of code (loc) in the file 10% -> 15%
- Cyclomatic Complexity 10% -> 17.5%
- Number of contributors (Potential)
- Branch coverage (Potential)

```
risk_score = (.225 * r) + ( .3 * f ) + ( .15 * m ) + ( .175 * c ) + ( .15 * loc )
```

Testing

- Tested REAL (normalized and fixed parameters)calculate_risk()
- Compare results with SampleDetector and Ground Truth
- Tested other sample repos

Top 3 Risky Files:

```
neuralop/training/trainer.py -> 0.80  
neuralop/models/fno.py -> 0.65  
neuralop/layers/spectral_convolution.py -> 0.65
```

trainer.py, local_no_block.py, spectral_convolution.py

neuralop/layers/local_fno_block.py -> 0.46

Risk Scores (File Name -> Risk Value):

```
modules/ui.py -> 0.87  
modules/metadata.py -> 0.22  
modules/processors/frame/face_swapper.py -> 0.35  
modules/globals.py -> 0.26  
modules/processors/frame/core.py -> 0.08  
modules/core.py -> 0.51  
modules/predictor.py -> 0.09  
modules/capturer.py -> 0.09  
modules/processors/frame/face_enhancer.py -> 0.27  
modules/face_analyser.py -> 0.39
```

Top 3 Risky Files:

```
modules/ui.py -> 0.87  
modules/core.py -> 0.51  
modules/face_analyser.py -> 0.39
```

Top 3 Risky Files:

```
- face_swapper.py -> 2.0  
- face_analyser.py -> 2.9103830456733704e-11  
- ui.py -> 1.45519159339652e-11
```

risky_files

face_swapper.py, ui.py, core.py

Top 3 Risky Files:

```
whisper/transcribe.py -> 0.76  
whisper/decoding.py -> 0.58  
whisper/timing.py -> 0.44
```

transcribe.py, timing.py, utils.py

whisper/utils.py -> 0.40

What's next?

- Entering Sprint 5
 - Collect feedback and apply fixes
 - Research more useful parameters
 - Implement the potential parameters
 - Start one page report
 - Documentation